Adaptive Confidence-Weighted LLM Infusion for Financial Reinforcement Learning

Emran Y. Alturki
Imperial College London
e.alturki24@imperial.ac.uk

Aydin Javadov
ETH Zürich
ajavadov@ethz.ch

Qiyang Sun
Imperial College London
q.sun23@imperial.ac.uk

Björn W. Schuller *Fellow, IEEE*
Imperial College London
Technical University of Munich
schuller@tum.de

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***Abstract*—We propose an enhanced version of FinRL-DeepSeek for the "FinRL Contest 2025 Task 1", a framework that integrates Large Language Model (LLM) signals into Reinforcement Learning (RL) for stock trading. The original approach employed fixed LLM-based perturbations (e. g., on actions or trajectory returns), overlooking each signal's uncertainty and thereby risking the amplification of noise.**

To address these limitations, we introduce *Adaptive Confidence-Weighted LLM Infusion* for two types of infusions. First, trading-actions scores, which aligns each LLM-derived factor around a neutral baseline based on the LLM's self-reported confidence. High-confidence signals preserve their strength, while low-confidence signals are dampened. Second, controlling the exploration-exploitation trade-off by entropy regularisation.This dynamic adaptation is implemented in both PPO and CVaR-PPO, enabling more robust, context-aware policy updates. Empirical backtests on real market data demonstrate that our method improves stability and risk-adjusted returns, underscoring the value of explicitly modeling confidence in LLM-augmented RL trading strategies. [1,2]

***Index Terms*—FinRL, Reinforcement Learning, Prompting, LLM Infusion, Financial Trading, Algorithmic Portfolio Management**

I. Introduction

Algorithmic trading is a rapidly evolving domain, and Reinforcement Learning (RL) has emerged as a powerful approach within it. However, as highlighted by [1] [2], there remains a notable gap in effectively integrating text-based market information (e. g., financial news) and addressing risk management in RL-driven trading strategies. This paper, produced under the guidelines of the FinRL Contest 2025, serves as a complementary extension of the aforementioned work, with an emphasis on refining prompt design, LLM infusion, and hyperparameter tunings. By doing so, we aim to enhance the robustness and performance of RL-based trading agents, ultimately laying a stronger foundation for future developments in algorithmic trading research.

II. Data and Methodology

In this section, dataset, LLM prompts, and algorithms will be discussed. The new LLM-infusion method will be discussed in a separate section.

[1]https://github.com/emranyat/FinRL_DeepSeek
[2]https://huggingface.co/datasets/emranyat/FinRL_2025

A. Datasets

Main datasets include Yahoo Finance for the Nasdaq-100 stocks information (e. g., prices, volumes, indicators, etc) [3]. Additionally, the FINSPID dataset [4] which was published last year and contains over 15.7 Million news on the Nasdaq-100 stocks, is used. This dataset was shortened randomly by [2] to 2 Million news for the purpose of LLM requests expenses.

B. LLM Prompts

- **Prompting Method:** According to [5], effective prompting for Large Language Models (LLMs) often involves a combination of strategies, such as *probscore*, *advanced descriptions*, and *few-shot learning*.
  1) *Probscore* gauges how certain the LLM is about its own response.
  2) *Advanced descriptions* ensure thorough background context for the LLM, potentially improving answer quality.
  3) *Few-shot learning* supplies a small number of examples (e. g., five), guiding the LLM's outputs via *In-Context Learning (ICL)*.
  4) *Date Specification* may influence an LLM to disregard knowledge acquired after the date a news item was published, reducing the risk of look-ahead bias on the score's evaluation.

  Based on [6], properly engineered prompts can outperform naive or purely human-crafted queries. Consequently, we utilize a highly ranked LLM (GROK-3 [7]) [8] to generate our final prompt (see Figure 1 for the risk score example), then feed that prompt into another LLM (DeepSeek-V3 [9]) to obtain the final scores. This same approach also applies to sentiment scoring, using a prompt closely resembling Figure 1.
- **Confidence Probabilities:** As described above, we request the LLM to output a confidence probability (or integer rating) alongside each score, recognizing that not all news items carry equal significance. Leveraging this confidence factor in our RL training (see Section III) helps the agent weigh uncertain signals more cautiously and trust highly confident signals more strongly.

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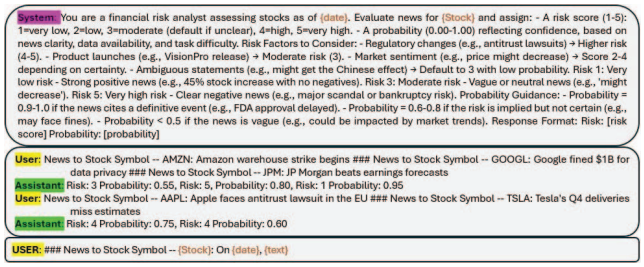

Fig. 1. Prompt for Risk Scores and Confidence Probabilities

C. Algorithms

We retain the two primary algorithms from the original paper [2], but enhance their integration with LLM signals:

a) Proximal Policy Optimization (PPO).: Originally introduced by OpenAI [10], PPO updates its policy by clipping the probability ratios between the new and old policies. This clipping mechanism penalizes excessive deviations, preventing unstable policy swings, and is typically yielding more robust performance in practice.

b) Conditional Value at Risk-Proximal Policy Optimization (CVaR-PPO): Proposed by [11], CVaR-PPO extends PPO by incorporating *risk-sensitive* optimization through Conditional Value at Risk (CVaR). This additional risk constraint helps mitigate large losses over adverse market conditions, often producing safer trading strategies than vanilla PPO. In our context, CVaR-PPO also benefits from the LLM-based signals, allowing us to more effectively manage and adapt to shifting risk profiles in the market.

III. Adaptive Confidence-Weighted LLM Infusion

In the original version of FinRL-DeepSeek [2], the LLM signals were injected into two main components of the RL agent:

1) A *recommendation factor* S_f, which amplifies or dampens each action a_t (e. g., pushing the policy to buy or sell).
2) A *risk factor* R_f, which modifies the trajectory returns $D(\pi_\theta)$ for risk-sensitive optimization (e. g., in CVaR-PPO).

However, both S_f and R_f were *fixed, discrete perturbations* close to 1.0, ignoring any confidence or uncertainty from the LLM output. In this section, we propose an **Adaptive Confidence-Weighted LLM Infusion** framework that accounts for the LLM's self-reported certainty, thereby yielding more robust and context-sensitive trading decisions.

A. Original Method: Fixed LLM Perturbations

In FinRL-DeepSeek, the recommendation factor S_f for each stock was chosen as a small multiplier in $\{0.9, 0.95, 1.0, 1.05, 1.1\}$ depending on the integer LLM sentiment score (1–5). Formally, the action became

$$a_t^{\text{mod}} \;=\; S_f \,\cdot\, a_t\,, \tag{1}$$

where S_f is close to 1.0. Meanwhile, the *risk factor* R_f was computed as a weighted sum across stocks,

$$R_f \;=\; \sum_i w_i\, R_f^i\,, \tag{2}$$

and used to adjust the trajectory returns for risk-sensitive optimization:

$$D_{R_f}(\pi_\theta) \;=\; R_f \;\cdot\; D(\pi_\theta)\,. \tag{3}$$

These S_f, R_f values were used as is per LLM score. While straightforward, the model does not distinguish between high-certainty and low-certainty LLM outputs, it is potentially amplifying noisy or unreliable signals.

B. Adaptive Confidence-Weighted Infusion

To incorporate the LLM's confidence explicitly, we propose a novel mechanism where each discrete LLM score (1–5) is accompanied by a confidence probability $C \in [0.00, 1.00]$ (where 0.00 indicates no confidence and 1.00 indicates a certain confidence in the score). We then replace the fixed factors with:

a) Adaptive Recommendation Factor: We define

$$\begin{aligned} S_f^{\text{mod}} &= 1 + (S_f - 1) * C_{\text{rec}} \cdot \alpha \\ a_t^{\text{mod}} &= S_f^{\text{mod}} \;\cdot\; a_t\,, \end{aligned} \tag{4}$$

where S_f is the *original* baseline factor from the LLM (e. g., $0.9, 0.95, 1.0, 1.05, 1.1$), and C_{rec} is the normalized confidence (ranging 0.00–1.00). The parameter $\alpha > 0$ is a hyperparameter controlling how strongly we amplify or dampen the recommendation based on confidence.

b) Adaptive Risk Factor: Similarly, for the risk component R_f^i, we introduce:

$$\begin{aligned} (R_f^i)^{\text{mod}} &= 1 + \left(R_f^i - 1\right) * C_{\text{risk}}^i \cdot \beta \\ R_f^{\text{mod}} &= \sum_i w_i \cdot \left(R_f^i\right)^{\text{mod}}\,, \end{aligned} \tag{5}$$

where each R_f^i is the baseline around 1.0, β is a tunable scaling parameter, and $C_{\text{risk}}^i \in [0.00, 1.00]$ is the confidence for each stock's risk assessment. The modified factor R_f^{mod} then replaces (2) in the CVaR or PPO objective:

$$D_{R_f^{\text{mod}}}(\pi_\theta) \;=\; R_f^{\text{mod}} \;\cdot\; D(\pi_\theta). \tag{6}$$

C. Why This Matters

The Adaptive Confidence-Weighted framework solves two critical issues:

1) **Noise Reduction**: In markets with frequent news and high data variability, ignoring confidence can overreact to spurious signals. Our confidence scaling dampens low-certainty updates.
2) **Improved Explainability**: Explicit confidence scores add interpretability to the RL agent's decisions. A strong recommendation with low confidence is visibly downgraded, preventing blind trust in questionable signals. The tuned parameters α and β also have interpretable meanings of 'how much the confidence score of the

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LLM' takes part in either recommendation or risk factor generation.

Moreover, the approach is modular. It requires modifications only to the final action (or risk) scaling, preserving the rest of the PPO/CPPO pipeline. Empirical results in Section IV demonstrate enhanced stability and higher risk-adjusted returns under volatile market conditions.

D. Implementation Remarks

Implementation involves three steps:

- **Prompting the LLM for Confidence**: Modify the LLM prompts to produce a confidence probability (C_{rec}, C_{risk}) in the range of $[0.00, 1.00]$.
- **Data Integration**: Append these confidence columns to the state representation for each stock and each time step.
- **Infusion Formulas**: Replace the original static multipliers S_f, R_f^i in (1)–(2) with the adaptive formulas (4)–(5) during each environment step.

This adjusted logic allows FinRL-DeepSeek to contribute to the trustworthiness of the LLM's news-driven signals by dynamically adapting to LLM confidence and without overhauling the underlying RL architecture.

E. Entropy Regularisation

This is a different LLM infusion method, where the entropy function is injected with the LLM scores. The purpose of this method is to control the trade-off between exploration-exploitation of the policy, where if a stock got a high recommendation score, the policy will exploit this option and opposite if the score is low, and vice versa for risk scores.

a) Recommendation Scores: The policy loss in PPO with entropy regularization is given by:

$$\mathcal{L}_\pi = -\,\mathbb{E}_t\left[\min\left(\frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)}A_t,\ \text{clip}\left(\frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)}, 1-\epsilon, 1+\epsilon\right)A_t\right)\right] + \beta\cdot\mathcal{H}(\pi_\theta(\cdot|s)) \tag{7}$$

where:

- $\mathcal{H}(\pi_\theta(\cdot|s)) = -\sum_a \pi_\theta(a|s)\log\pi_\theta(a|s)$ is the **entropy**.
- β is the entropy coefficient (hyperparameter).

where the function of β is one of two options:

$$\beta = coef\cdot(5 - S_f)\cdot C_{\text{rec}} \tag{8}$$

Or;

$$\beta = S_f\cdot C_{\text{rec}}^2 \tag{9}$$

where S_{LLM} is the recommendation score that was assigned by the LLM (1-5).

b) Risk Scores: In the case of risk scores in CPPO:

$$\beta = coef\cdot R_f\cdot C_{\text{risk}} \tag{10}$$

Or;

$$\beta = R_f\cdot C_{\text{risk}}^2 \tag{11}$$

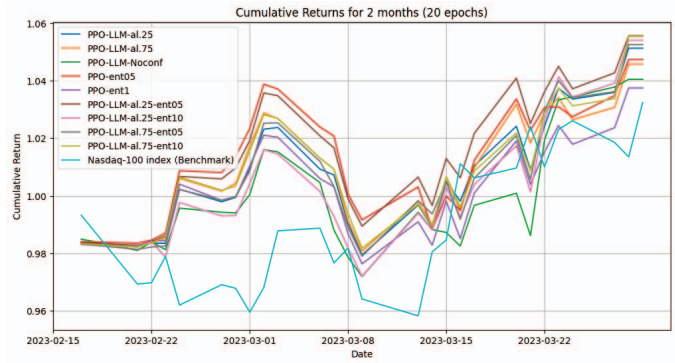

Fig. 2. Backtesting of the highest returns PPO models after 400K training steps (20 epochs, 20k steps), 14 months training, 2 months trading

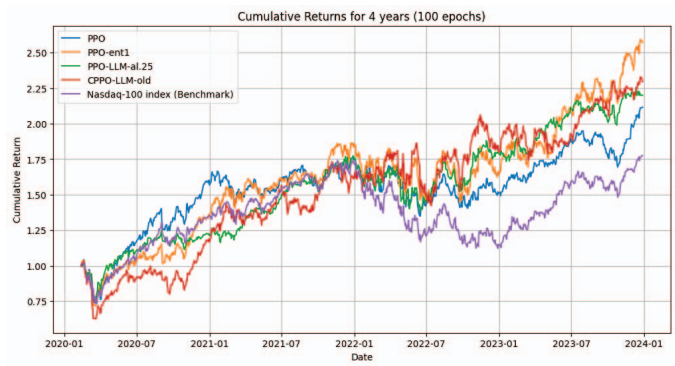

Fig. 3. Backtesting of the best models (ours compared to the old paper [2]) after 2M training steps (100 epochs, 20k steps), 6 years training, 4 years trading

F. Adaptive Factors + Entropy Regularization

We propose the merging of both LLM infusion methods, where injection will occur at the policy level to amplify / damage the action and at the entropy level to mitigate the trade-off between exploration and exploitation.

IV. Results

A. Early Stopping

Experiments were conducted to a small subset of the whole dataset (2013-2023) to reduce computational costs and explore different pathways. Specifically, we tried a 10% amplifier/damper of the score for both PPO and CPPO.

Training Period: 2022 - 2023

Trading Period: 2023/2 - 2023/3

Figure 2 and table I present the preliminary results of early stopping (400k training steps). Results indicate that the use of CPPO with both LLm infusions (factors and entropy) got a higher cumulative returns than the Nasdaq-100 benchmark, indicating that the infusion of recommendation/risk scores and its confidences have positively contributed to the model.

B. Full Training

Among the models tested in the early stopping stage, several models have been chosen to apply the full training; owing

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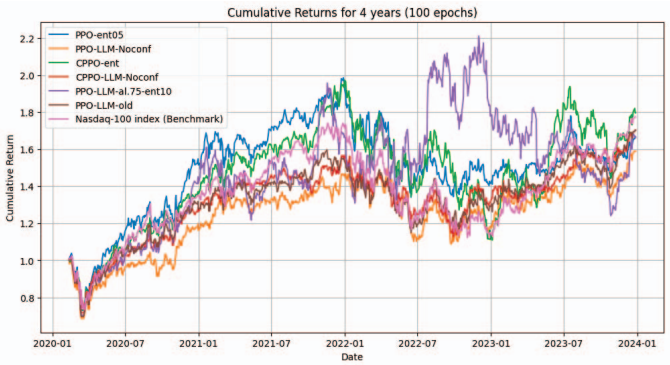

Fig. 4. Backtesting of several models after 2M training steps (100 epochs, 20k steps), 6 years training, 4 years trading

TABLE I
BACKTESTING RESULTS (20 EPOCHS)

Model	Cumulative Returns	Rachev Ratio	Max Drawdown	Outperf. Frequency	Downturn Outperf.
PPO	-0.0139	1.1459	-0.0697	40.7407	66.6667
PPO-LLM al.25	0.0513	1.1095	-0.0206	55.5556	91.6667
PPO-LLM al.5	0.0306	1.2579	-0.0250	55.5556	91.6667
PPO-LLM al.75	0.0458	1.2677	-0.0192	55.5556	91.6667
PPO-LLM Noconf	0.0405	1.7968	-0.0281	51.8519	83.3333
PPO-ent05	0.0473	1.2053	-0.0165	55.5556	91.6667
PPO-ent1	0.0375	1.1663	-0.0237	55.5556	91.6667
PPO-ent10	-0.0110	0.8529	-0.0544	51.8519	83.3333
PPO-ent	0.0209	1.0510	-0.0510	55.5556	83.3333
PPO-LLM al.25 & ent05	0.0556	1.1806	-0.0173	55.5556	91.6667
PPO-LLM al.25 & ent10	0.0541	1.4350	-0.0280	59.2593	91.6667
PPO-LLM al.75 & ent05	0.0526	1.3778	-0.0210	55.5556	91.6667
PPO-LLM al.75 & ent10	0.0554	1.3146	-0.0182	55.5556	91.6667
CPPO-LLM al.25-be.25	-0.0073	0.7725	-0.0463	48.1481	83.3333
CPPO-LLM al.25-be.5	-0.0087	1.0660	-0.0566	44.4444	83.3333
CPPO-LLM al.25-be.75	0.0057	0.9999	-0.0421	48.1481	83.3333
CPPO-LLM al.5-be.25	0.0011	0.9517	-0.0431	48.1481	83.3333
CPPO-LLM al.5-be.5	-0.0039	0.9040	-0.0394	48.1481	83.3333
CPPO-LLM al.5-be.75	0.0148	1.1834	-0.0449	51.8519	83.3333
CPPO-LLM al.75-be.25	0.0277	1.0334	-0.0312	51.8519	83.3333
CPPO-LLM al.75-be.5	0.0051	0.9088	-0.0447	48.1481	83.3333
CPPO-LLM al.75-be.75	-0.0059	0.8086	-0.0486	48.1481	83.3333
CPPO-LLM Noconf	0.0272	1.2135	-0.0223	51.8519	91.6667
CPPO-ent05	-0.0008	0.9642	-0.0409	48.1481	83.3333
CPPO-ent1	0.0206	1.2660	-0.0378	51.8519	83.3333
CPPO-ent10	0.0033	0.9216	-0.0451	51.8519	83.3333
CPPO-ent	0.0136	1.0620	-0.0355	55.5556	91.6667
CPPO-LLM & ent1	0.0269	1.2671	-0.0315	51.8519	83.3333

to limited computational resources. The complete training includes the entire dataset (2013-2023), which is divided as follows:

Training Period: 2013 - 2018

Trading Period: 2019 - 2023

Figure 3, 4 and table II show the results after (2 Million training steps). Results show that our methods got higher returns especially with PPO model. When comparing the winner prposed model (PPO-ent1) with the winner model of

TABLE II
BACKTESTING RESULTS (100 EPOCHS)

Model	Cumulative Returns	Rachev Ratio	Max Drawdown	Outperf. Frequency	Downturn Outperf.
PPO-ent1	1.5715	1.0367	-0.2807	48.7701	74.3468
CPPO-LLM (old)	1.2960	0.9698	-0.3742	48.7701	70.3088
PPO	1.1156	0.9982	-0.2571	48.7701	79.3349
PPO-LLM al0.25	1.1996	0.9567	-0.2419	47.2727	78.8599
CPPO-ent	0.7969	1.0339	-0.2676	49.4118	74.1093
CPPO-LLM-Noconf	0.7033	0.9664	-0.2846	48.4492	79.8100
PPO-LLM (old)	0.7031	0.9426	-0.3028	50.5882	78.8599
PPO-ent05	0.6665	0.9492	-0.2694	50.4813	77.1971
PPO-LLM al0.75 & ent10	0.6623	1.0413	-0.2968	50.1604	72.6841
PPO-LLM-Noconf	0.5887	0.9673	-0.3133	47.3797	74.3468

the previous paper (CPPO-LLM (old)), it is noticed that our model got better returns with generally less risks associated.

V. CONCLUSION

We introduced an enhanced version of FinRL-DeepSeek by refining its LLM prompt design and incorporating an adaptive LLM-infusion approach. Empirical evidence suggests that our improvements can significantly stabilize performance and better leverage text-based financial information for PPO models especially. In future work, we aim to:

- **Expand News Sources:** Incorporate diverse media (e. g., X-posts, YouTube videos) to capture a broader market sentiment spectrum and enable more robust decision-making.
- **Pipeline Optimization:** Streamline the data processing and request management pipeline for LLM interactions. In our initial trials, parallelizing the requests to the LLM provider reduced average latency by up to 90%, minimizing dropped queries and missing data.

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